

AI/ML Foundations - Lab Assignment – M1

End-to-End ML Mini Project

Credit Default Risk Prediction

Total Marks: 50 Dataset: Credit Default dataset (e.g., UCI Credit Card Default or similar structured dataset)

Target: default (1 = default, 0 = no default)

Objective

Build a complete ML pipeline to predict the probability of loan default and demonstrate:

- Understanding of the ML lifecycle
- Rigorous data preparation
- Baseline comparison
- Metric trade-offs and threshold tuning
- Offline vs Online reasoning
- Awareness of production deployment and monitoring

Part 1 - Define & Feasibility (5 Marks)

Tasks:

1. Write a clear problem statement (2–3 lines).
2. Define:
 - Target variable (binary: default vs no default)
 - Offline success metric (e.g., F1-score, Recall)
 - Business metric (online, e.g., cost savings, reduced defaults, operational efficiency)
3. Create a simple baseline rule.
 - Example: If credit utilization > 80%, predict default.
4. Evaluate baseline performance using metrics (Accuracy, Precision, Recall, F1).

Hints:

- Accuracy alone is misleading due to class imbalance.
- Default detection often prioritizes Recall (catching defaulters).

Marks:

- Problem clarity – 2
- Metric reasoning – 1
- Baseline implemented – 2

Part 2 - Data Preparation (10 Marks)**Tasks:**

1. Perform train/test split (use stratified sampling to preserve class distribution).
2. Handle missing values (imputation strategies: mean/median for numeric, mode for categorical).
3. Encode categorical variables (One-Hot Encoding or Label Encoding).
4. Scale numerical variables.
5. Ensure reproducibility with a fixed random seed.
6. Implement all steps in a Pipeline for clean workflow.

Marks:

- Proper split (stratified) – 2
- Missing value handling – 3
- Encoding + scaling – 3
- Reproducibility – 2

Part 3 - Train Baseline Model (5 Marks)**Tasks:**

1. Train a Logistic Regression model.
2. Report metrics: Accuracy, Precision, Recall, F1.
3. Show and interpret the confusion matrix.

Important:

- Highlight how class imbalance affects metrics.
- Discuss why Recall might be more important than Precision in this context.

Marks:

- Model training – 2
- Metrics reported – 2
- Confusion matrix interpreted – 1

Part 4 - Threshold Trade-off Analysis (8 Marks)

Scenario: Business says: “Missing a defaulter is very costly, but too many false alarms increase operational cost.”

Tasks:

1. Compare thresholds = 0.5, 0.3, 0.7.
2. Show how Precision and Recall change at each threshold.
3. Write explanation:
 - Which threshold would you recommend?
 - Why?

Marks:

- Correct threshold logic – 3
- Metric comparison – 3
- Business-aligned reasoning – 2

Part 5 - Error Analysis (5 Marks)**Tasks:**

1. Analyze False Negatives (missed defaulters).
2. Identify patterns (e.g., low income, certain age groups, high utilization).
3. Write 3–5 insights with supporting evidence.

Marks:

- Proper error extraction – 2
- Insightful reasoning – 3

Part 6 - Production & Monitoring Simulation (7 Marks)

Tasks (answer in markdown):

1. Define **input schema** (fields + types).
 - Example: income: float, age: int, job_category: string, credit_utilization: float
2. What happens if:
 - Income column missing?
 - New unseen job category appears?
3. What would you monitor in production?
 - Data drift (e.g., income distribution shifts)
 - Operational metrics (e.g., % flagged defaults, false alarms)
4. When would you retrain?
 - Example: If drift is detected, or performance drops below threshold.

Marks:

- Schema clarity – 2
- Validation thinking – 2
- Monitoring awareness – 2
- Retraining reasoning – 1

Part 7 - Offline vs Online Reflection (5 Marks)

Scenario: Model F1 improved from 0.72 → 0.80 offline. But loan approval rate dropped and customer complaints increased.

Tasks:

1. Explain why this happened (offline vs online mismatch).
 - Example: Model became stricter, rejecting more loans → fewer approvals, more complaints.
2. Give 2 online metrics for this system.

- Example: Loan approval rate, customer satisfaction score.

3. Complete sentence:

- Offline metrics tell you _____.
- Online metrics tell you _____.

Marks:

- Correct conceptual reasoning – 3
- Practical online metric examples – 2

Bonus (Optional +5 Marks)

Choose one:

- Add `class_weight='balanced'` in Logistic Regression and compare results.
- Train a Decision Tree and compare overfitting behavior.
- Simulate data drift (e.g., shift income distribution) and observe metric change.

Final Marking Summary

Component	Marks
Define & Feasibility	5
Data Preparation	10
Training & Metrics	5
Threshold Analysis	8
Error Analysis	5
Production Thinking	7
Offline vs Online	5
Total	50